ScopeX Advanced Oscilloscope

Setup Manual

Driver Installation & Host Software Framework

ScopeX Integrated Laboratory Framework

- Dual-Channel Digital Storage Oscilloscope (DSO)
- Integrated DDS Analog Signal & Digital Frequency Generator
- Fixed +5VDC and High-Precision Bipolar ±12V Dual Rail Supply
- Cross-Platform Native GUI Deployment (Windows, macOS, Linux)

Revision Tracking: v3.0

Supported Architecture Nodes: Windows 10/11 · macOS Sierra+ · Linux (Ubuntu 20.04+)

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1 Introduction

Overview: The Scope-X reference hardware system from Bumblebee Instruments relies on an onboard ATxmega processor executing a standard USB Communications Device Class (CDC-ACM) profile.
Note: This manual contains information regarding setup for Windows only. MacOS, Linux will be released in the next version.

This manual details the procedures required to flash, map, and bind host drivers to the Scope-X GUI interface application framework.

Depending on your structural target host platform (Windows, macOS, or Linux), device node acquisition requires specific local terminal access privileges. Users are advised to retrieve the structural workspace bundle containing the host executable binaries and the verbatim archive file named `Drivers.zip` before proceeding with initialization strings.

1.1 Cross-Platform Core Infrastructure

- **Windows 10/11 Deployment:** Utilizes inf-driven virtual structural mapping to allocate system terminal objects.
- **macOS Architectures:** Operates via the built-in communication class system layer, utilizing Gatekeeper pathway authorization.
- **Linux Environments:** Relies on kernel-level `cdc_acm` modules requiring structural group permission modifications.

1.2 Kit Components Inventory

Table 1: System Hardware Packaging Profile

Component Description	Interface Specification	Quantity
Scope-X Main Processing Board	ATxmega Core / Analog Signal Mesh Layer	1
High-Speed Comm-Link Cable	Custom Shielded USB-B Link Cable	1

2 System Setup and Configuration

Important Deployment Rules: Observe clean Electrostatic Discharge (ESD) procedures when manipulating the processing architecture. Ensure the host processing terminal environment satisfies the cross-platform dependencies listed below.

2.1 Initial Physical Layout Logic

1. Unpack the structural hardware component stack and evaluate it against the packaging layout documentation.
2. Extract the software workspace folder onto a verified local storage matrix tracking route.
3. Provide the hardware block with proper power via the tracking cable. Verify active LED status loops.
4. Secure the USB-B link to target structural host ports to register internal endpoints.

Host Terminal System Requirements

- **Windows Platform:** Windows 10 / 11 (x64 Architecture)
- **Apple Platform:** Intel / Apple Silicon (M1/M2/M3) macOS 10.12+
- **Linux Open-Source:** Kernel 5.4+ (Ubuntu 20.04 LTS or equivalent)
- **Physical Port Allocation:** Dedicated USB 2.0/3.0 Type-A/Type-C Slot

2.2 Windows Setup Procedure

2.2.1 Install Application and Drivers

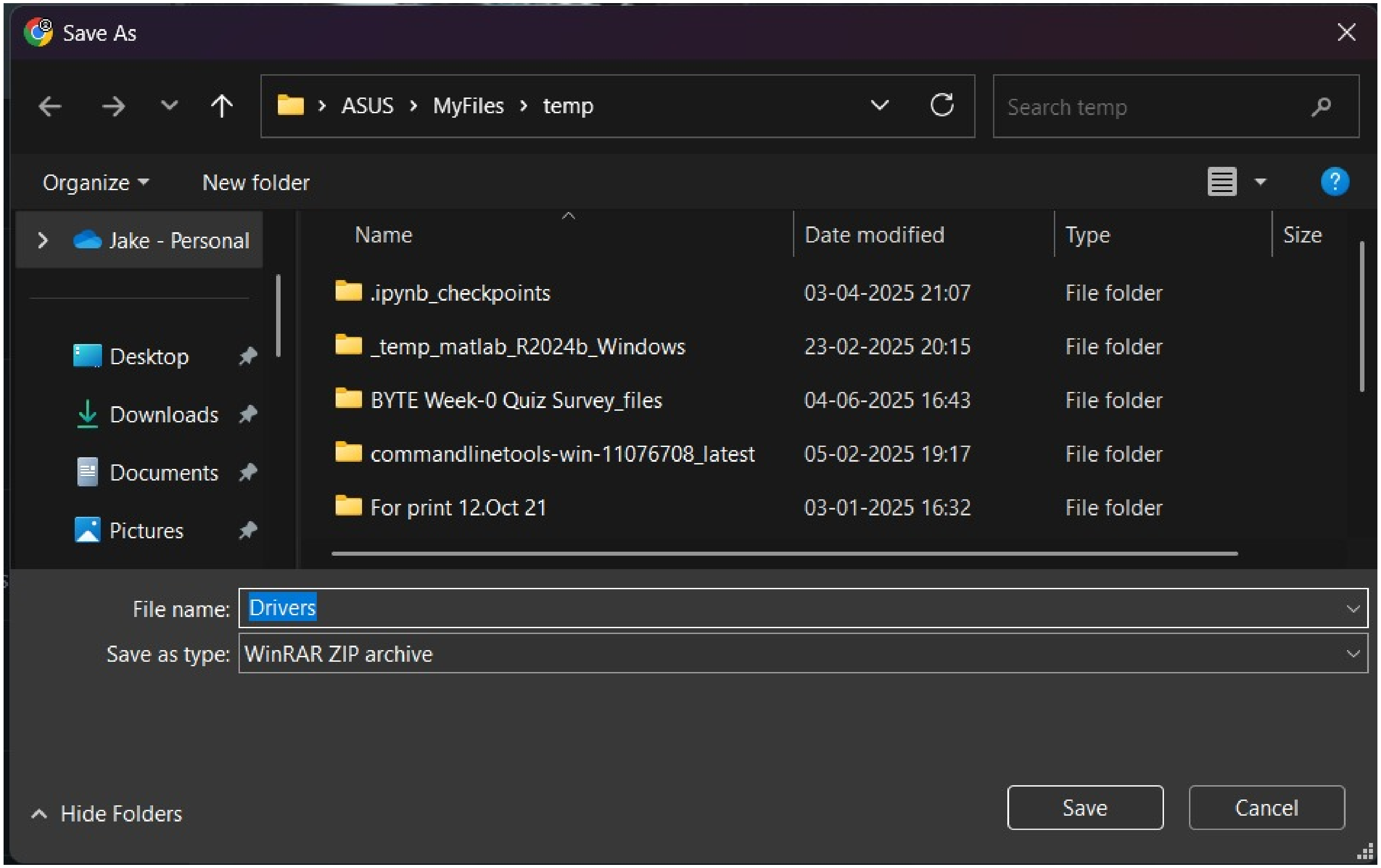

Figure 1: Locate your workspace bundle. Save the verbatim `Drivers.zip` archive and extract its contents to your preferred system directory alongside the main executable binary.

2.2.2 Open Executable

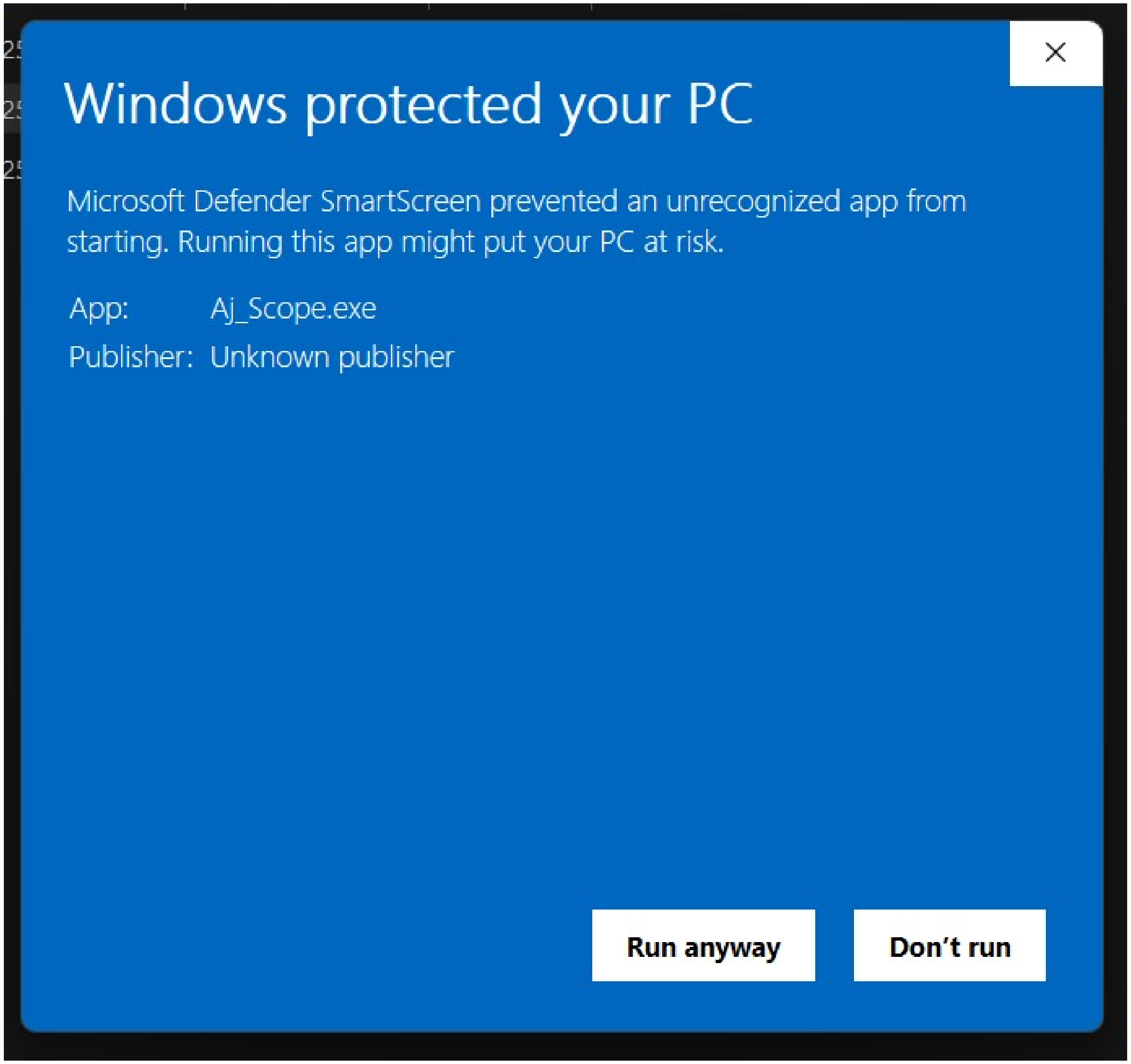

Figure 2: Open the host executable application. Select 'Run anyway' if prompted by the Windows SmartScreen security filter.

2.2.3 Connect Scopex and Find Device

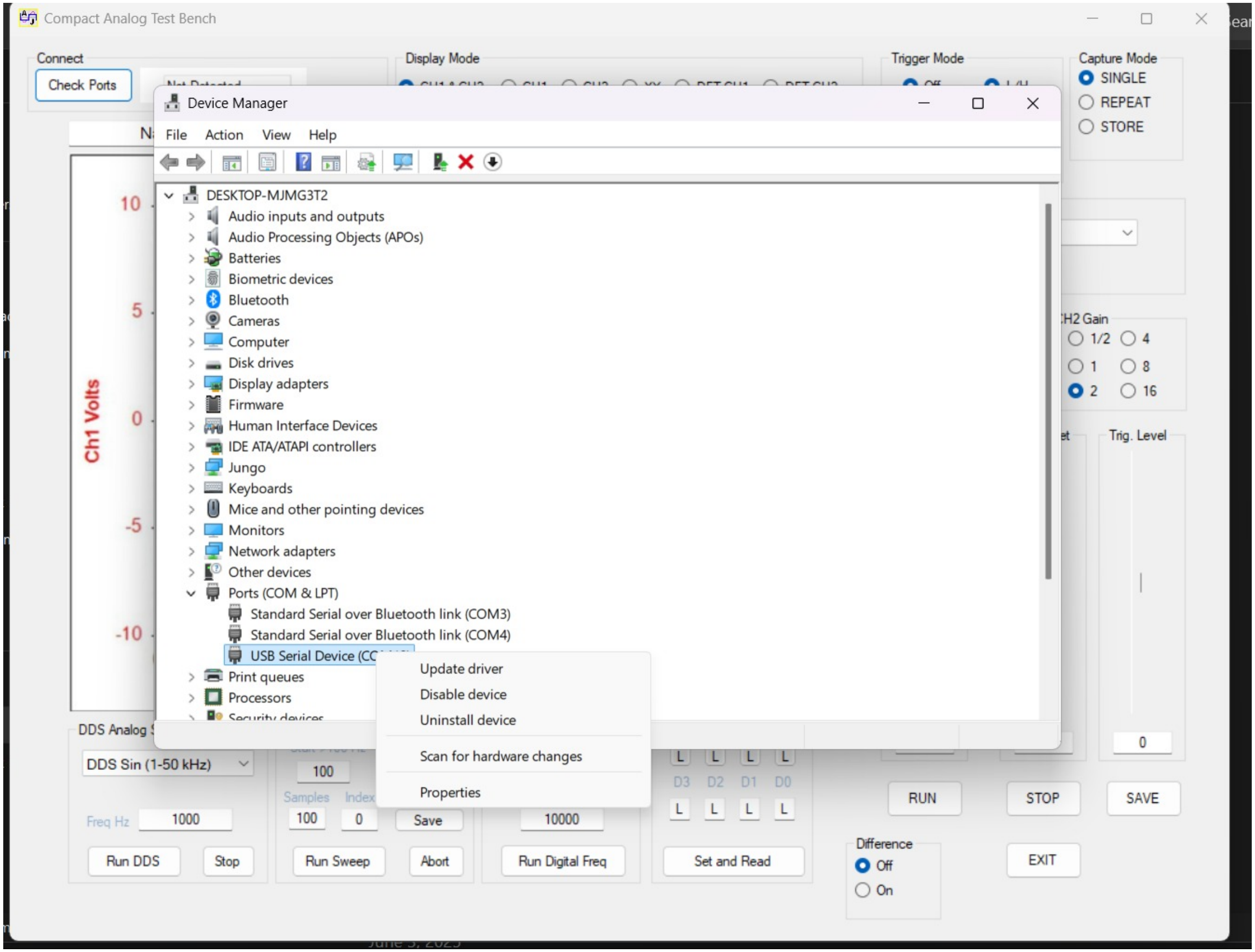

Figure 3: Interface the core board via USB. Open Device Manager and locate the unmapped 'USB Serial Device' entry inside the Ports (COM & LPT) categorization path.

2.2.4 Update Driver

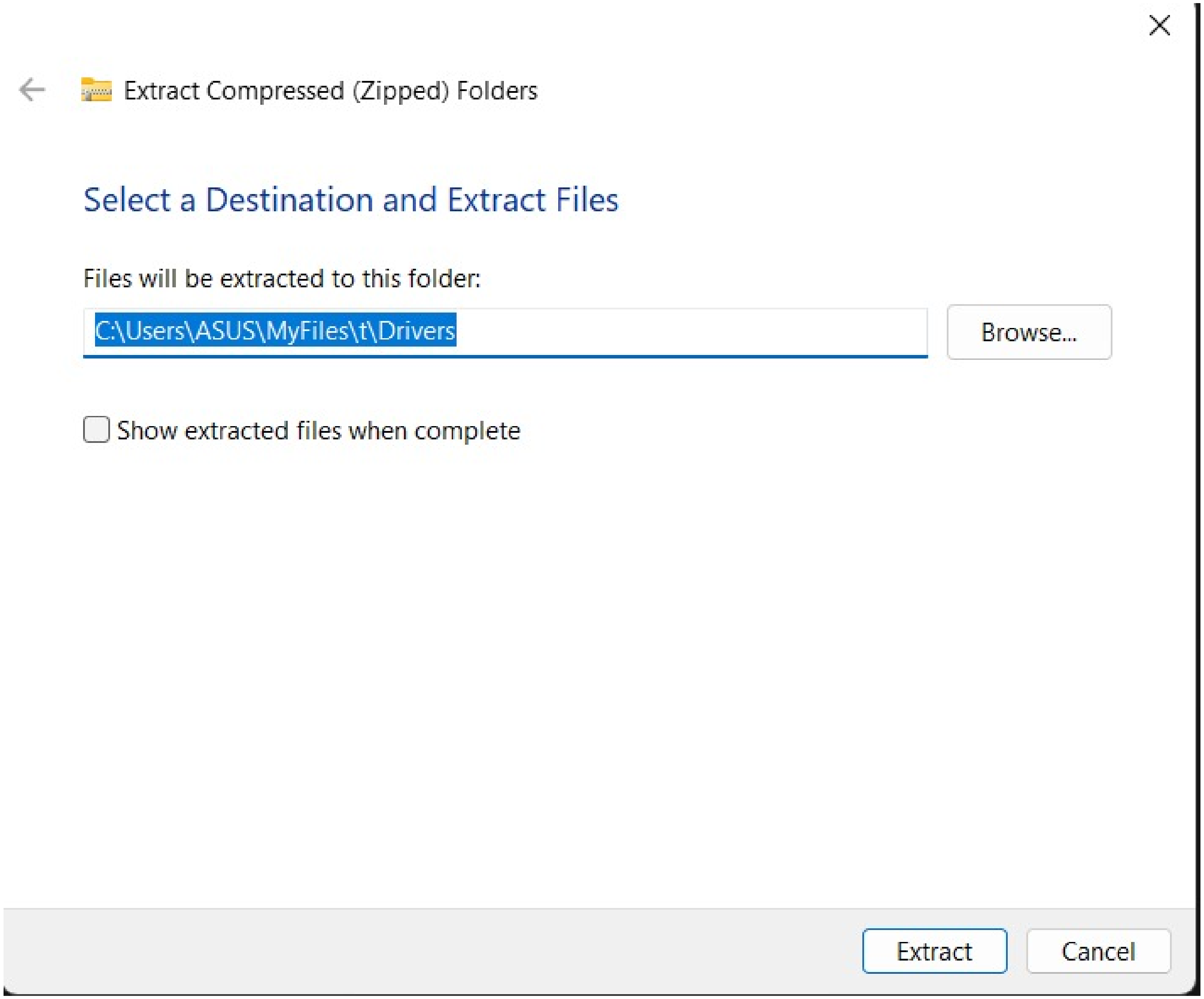

Figure 4: Right-click on the device entry, choose 'Update Driver', select the manual local lookup parameter, and navigate to the directory containing the uncompressed structure from `Drivers.zip`.

2.2.5 Install Driver

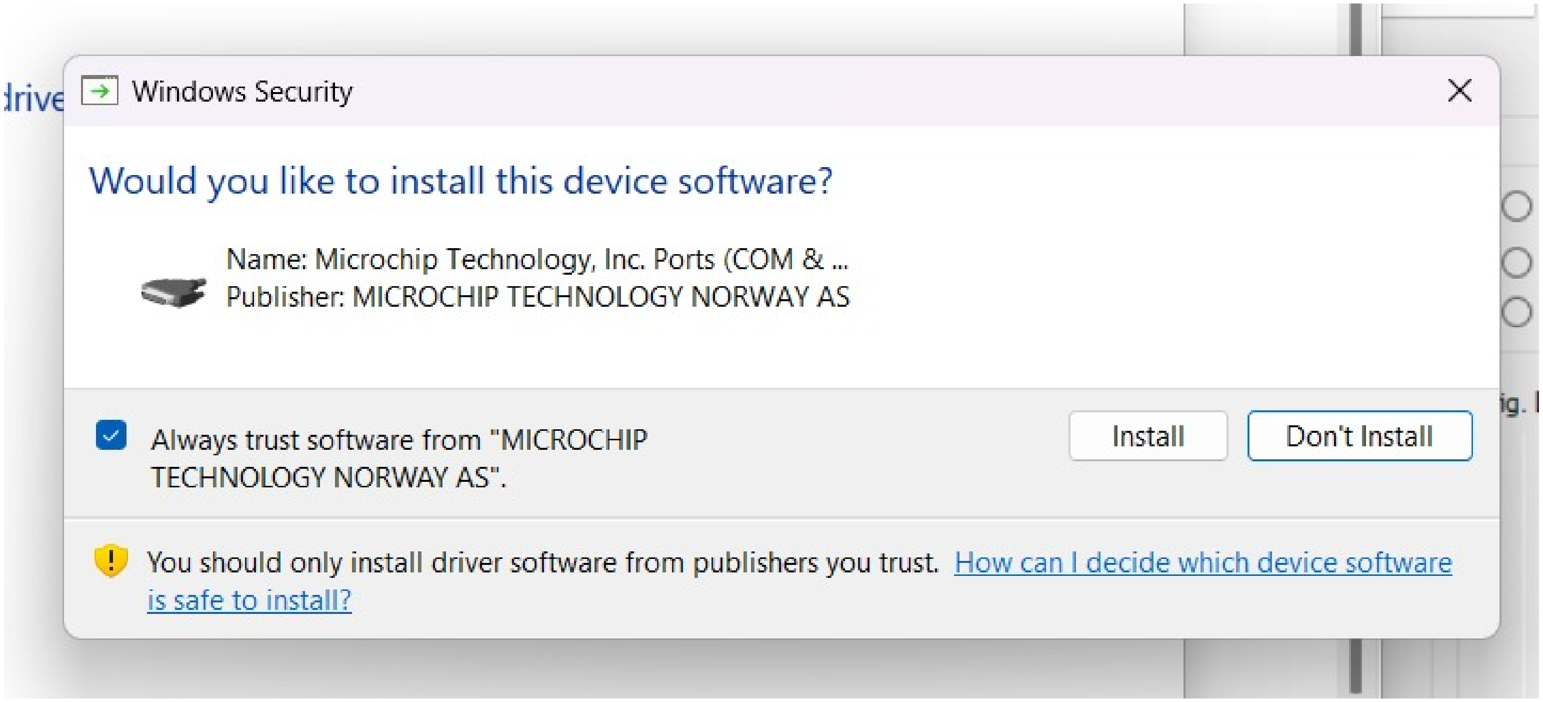

Figure 5: Accept any system prompts to authorize initialization. The tracking engine registers the inf parameters into host infrastructure indexes.

2.2.6 Verify COM Port Recognition

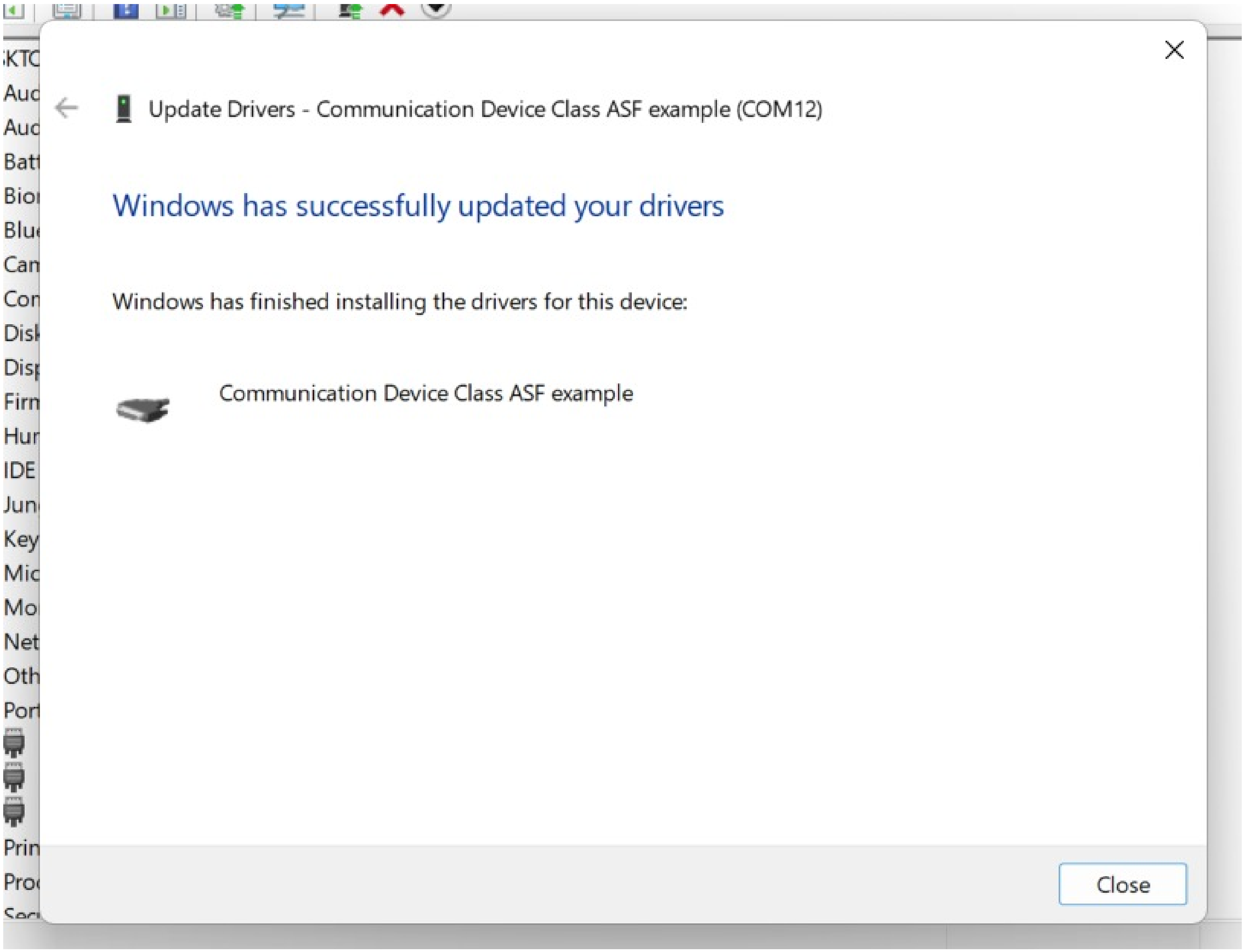

Figure 6: Verify that the system architecture successfully maps the target hardware, re-indexing it as a verified virtual communication endpoint node under an assigned COM identification block.

2.2.7 Launch GUI and Check

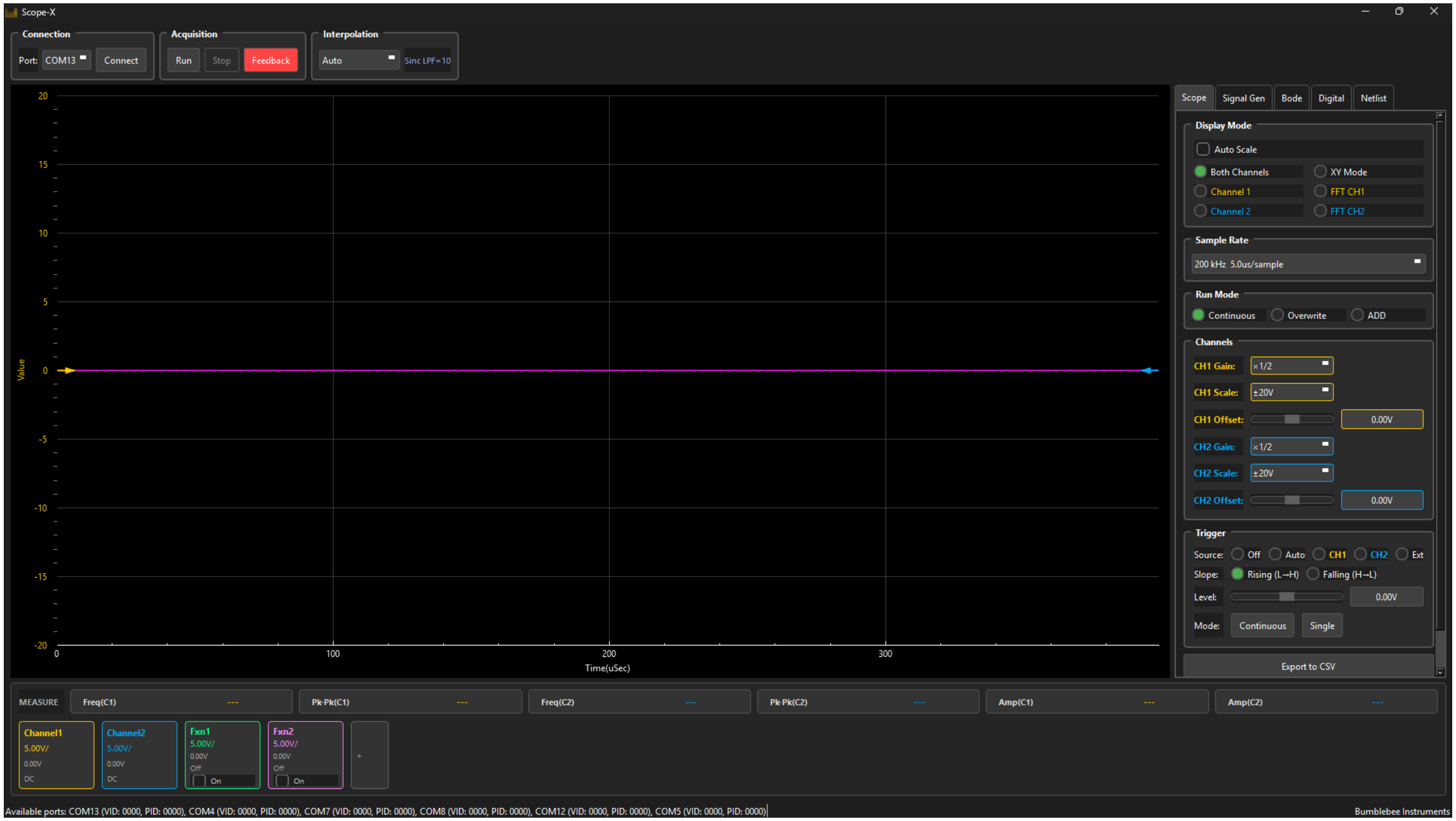

Figure 7: Once you launch the GUI, it tries to auto connect to the device, else you can try to Connect and now the device is connected to the GUI

3 Hardware Architecture

3.1 Main Board Overview

The Scope-X main board features a modular architecture optimized for educational use and rapid prototyping.

Figure 8: Scope-X Main Board

3.2 Board Features

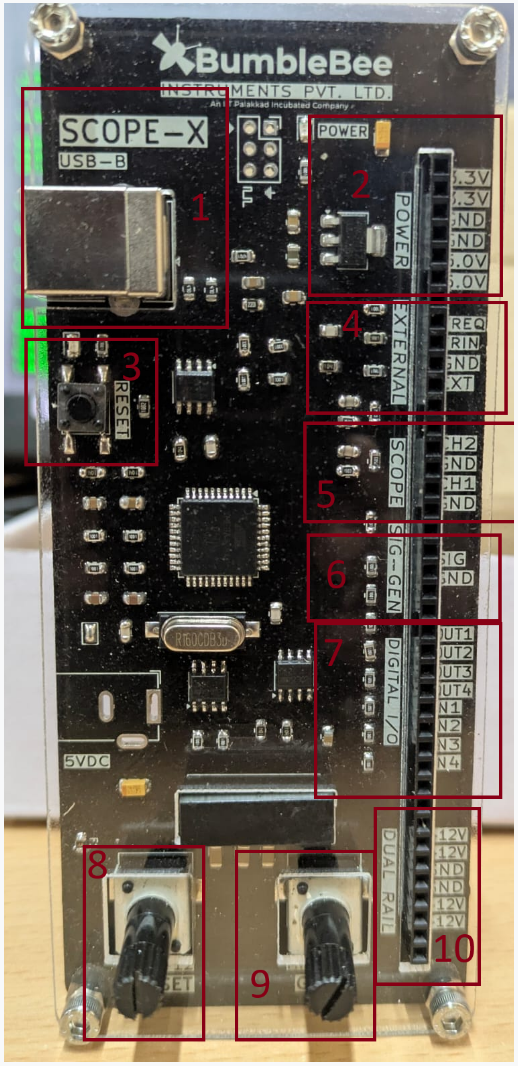

Figure 9: Scope-X Processing Hardware Subsystem

Table 2: System Mapping Allocation Table

Feature Assignment	Operational Use Case Parameters
1 USB-B Port	High-Speed Virtual COM Pipeline
2 Power Switch	Main Rail Isolation Management
3 Reset Link	Core Microcontroller Restart Matrix
4 External Bus	Accessory Processing Routing Matrix
5 Scope Inputs	Dual-Channel System Diagnostics Block
6 DDS Generator	Programmable Continuous Waveform Matrix
7 Digital I/O Block	4-Bit Capture and Stimulation Channels
8 Offset Adjust	Operational Tuning Block (-12V to +12V)
9 Gain Matrix	Precision Hardware Scaler Interface
10 Dual Rail Block	Bipolar Lab Reference Subsystem